Definition 7.2

A random variable X is said to follow a Poission distribution with parameter λ if it assumes only non–negative values and its probability mass function is given by

$$P(x,\lambda) = P(X=x) = \begin{cases} \dfrac{e^{-\lambda}\lambda^{x}}{x!}, & x = 0,1,2\ldots\ldots;\lambda>0 \\ 0 & otherwise \end{cases}$$

Derivation of Mean and variance of Poisson distribution

$$\text{Mean } E(X) = \sum_{x=0}^{\infty} x\,p(x,\lambda)$$

$$= \sum_{x=0}^{\infty} x\frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$= \lambda e^{-\lambda}\left\{\sum_{x=1}^{\infty}\left(\frac{\lambda^{x-1}}{(x-1)!}\right)\right\}$$

$$= \lambda e^{-\lambda}(1+\lambda+\lambda^2/2!+\ldots\ldots)$$

$$= \lambda e^{-\lambda} e^{\lambda}$$

$$= \lambda$$

$$\text{Variance } (X) = E(X^2) - [E(X)]^2$$

$$\text{Here } E(X^2) = \sum_{x=0}^{\infty} x^2 p(x,\lambda)$$

$$= \sum_{x=0}^{\infty} x^2 p(x,\lambda)$$

$$= \sum_{x=0}^{\infty}\{x(x-1)+x\}p(x,\lambda)$$

$$= \sum_{x=0}^{\infty}\{x(x-1)+x\}\frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$= e^{-\lambda}\sum_{x=0}^{\infty}x(x-1)\lambda^{x}/x! + \sum_{x=0}^{\infty} x\,e^{-\lambda}\frac{\lambda^{x}}{x!}$$

$$= \lambda^2 e^{-\lambda}\sum_{x=2}^{\infty}\frac{\lambda^{x-2}}{(x-2)!} + \lambda$$

$$= \lambda^2 e^{-\lambda} e^{\lambda} + \lambda = \lambda^2 + \lambda$$

$$\text{Variance } (X) = E(X^2) - [E(X)]^2$$

$$= \lambda^2 + \lambda - (\lambda)^2$$

$$= \lambda$$

Properties of Poisson distribution :

1. Poisson distribution is the only distribution in which the mean and variance are equal .

Example 7.14

In a Poisson distribution the first probability term is 0.2725. Find the next Probability term

Solution :

Given that $p(0) = 0.2725$

$$\frac{e^{-\lambda}\lambda^{0}}{0!} = 0.2725$$

$$\Rightarrow e^{-\lambda} = 0.2725$$

(by using exponent table)

$$\lambda = 1.3$$

$$\therefore p(X=1) = e^{-1.3}(1.3)/1!$$

$$= e^{-1.3}(1.3)$$

$$= 0.2725 \text{ x } 1.3$$

$$= 0.3543$$

Example 7.15

In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of errors per page, find the probability that a random sample of 5 pages will contain no error.

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